Testing, Acceptance & Anti-Bluff QA Strategy (constitution §11.4.169)

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> **THE ONE-SENTENCE RULE (plain engineering language, no jargon, read this first). A green test suite is not proof the feature works. It only counts as proof if a real artifact — a packet capture, a throughput number, a database rowset, a screen recording — was captured *while the feature ran* and shows the user-visible outcome actually happened.** "The assertion returned true" is not evidence. "Here is the pcap showing the SYN-ACK from the server the user was supposed to reach" is evidence. Every test type in this document (unit through security through UI) is built around this one rule; §0 below restates it as the formal doctrine and §3 fixes exactly what artifact each feature class must produce. If a test, a review, or a release checklist ever accepts "tests pass" on its own as done, that is a defect in the QA process itself, not an acceptable shortcut — it is precisely the failure mode this whole document exists to make impossible.
>
> **Rev 3 (2026-07-04, gap-analysis + hardening pass):** added this explicit plain-language restatement of the anti-bluff bar as the first thing a reader sees (previously the same substance existed only inside the §0 doctrine section, layered under constitution citations — this makes it unmissable and citable on its own). No other normative content changed in this pass; the taxonomy (§2), evidence model (§3), per-type harnesses (§5), coverage ledger (§6), and acceptance gates (§7) were independently re-verified

> against `SPECIFICATION.md` / `decision-register.md` and found internally consistent — no contradictions found requiring correction.

> **Reconciled (§11.4.35, 2026-06-26):** §2 now states the test-type set consistently as **16 bundled families / 18 enumerated codes** (resolving the prior "Sixteen types" prose vs the 18 `test_type` CHECK codes in the v08 coverage-ledger schema). §7.2 (DoD = nine ACs, AC1–AC9) is unchanged and is the canonical authority the v08 child docs were reconciled to.

> Master technical specification — document 10 of the HelixVPN set. This is the **canonical QA contract**: it defines, once, the §11.4.169 test-type taxonomy, the captured-evidence model, the harnesses, the per-phase acceptance gates, and the coverage-ledger schema that the Phase-0 and Phase-1 work-breakdown documents (06, 07) reference by abbreviation. Every other document's per-item `Tests:` column draws from the closed vocabulary fixed *here* (§2). It is **spec-only** — it describes *how* "*works for the end user*" *is mechanically proven*, never the product itself (2–3 refinement passes follow).
>
> Authoritative companions cited inline: data plane [01]; control plane [02]; client core + UI [03]; security/privacy/PKI [04]; repo + ecosystem [05]; Phase-0 WBS [06]; Phase-1 WBS [07]. Primary research: **[04_P0]** (`04_VPN_CLD/HelixVPN-Phase0-Spike.md`), **[04_P1]** (Phase1-MVP), **[04_ARCH]**, **[04_UI]** (`HelixVPN-helix-ui-Flutter.md`), **[05_YBO]**, **[SYNTHESIS]**, LLM analyses **[02_QWN] [01_DSK] [11_MST] [10_KMI] [07_GMI]**, external facts **[research-masque]** (RFC 9297/9298/9221), **[research-boringtun]**, **[research-quinn]**, **[research-wireguard]**.

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0. Doctrine: the anti-bluff covenant applied to a VPN

A VPN is the worst possible product to bluff-test, because **every failure mode is silent and invisible to a green summary line**. A kill-switch that "passes" but leaks one DNS query deanonymises the user. A tunnel that "connects" but routes plaintext is worse than no VPN. A policy edit that "applies" but leaves a stale `AllowedIPs` grants access that was revoked. The §11.4 covenant's founding mandate — "*execution of tests MUST guarantee … full usability by end users*" — is not a slogan here; it is the only thing standing between the product and a deanonymisation incident.

Therefore the operative bar for **every** HelixVPN test is not "the assertion returned true" but **"a packet, a pcap, a counter delta, or a rendered frame — captured during execution — proves the user-visible security property held."** Five HelixVPN-specific bluff classes are forbidden by name (each maps to a constitution clause and is mechanically guarded in §5):

Bluff class	What it looks like	Forbidden by	Guarded in
B1 — config-only PASS	asserting `AllowedIPs` contains the route, never sending a packet	§11.4.69	§5.3 E2E, §5.7 SEC
B2 — absence-of-error PASS	"kill-switch enabled, no error" with no leak-capture	§11.4.68 / .107	§5.7 SEC, AC7 (§7.2)

Bluff class	What it looks like	Forbidden by	Guarded in
B3 — wrong-plane PASS	`dumpsys`/policy field says "connected" while no PCM/ packet flows	§11.4.107(2), §11.4.68	§5.3, §5.13
B4 — stale-state PASS	validating against a previous deploy's running daemon	§11.4.108/.139	§8, §9
B5 — unvalidated-analyzer PASS	the pcap/OCR analyzer passes its own golden-bad fixture	§11.4.107(10)	§3.3, §5.14

Every PASS in this strategy flows through `ab_pass_with_evidence <desc> <evidence_path>` (§11.4.69); bare `ab_pass` is a release blocker. A FAIL that crashes for a script-internal reason (`set -u` undefined var, a broken `tshark` filter) is **equally forbidden** (§11.4.1) — the harness sources a shared anti-bluff lib (§9) so script bugs surface at the lib layer, not as fake product FAILs.

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1. §11.4.169 — the universal test-type mandate

§11.4.169 (the §11.4.27 enumeration, promoted to a per-item closure obligation) requires that **every workable item declares the closed set of test types it must pass before it may close**, and that the *absence* of any warranted type is never silent — it is an explicit `NOT_APPLICABLE: <reason>` or an honest §11.4.3 `SKIP: <closed-set-reason>`, never an omission. This document fixes that closed set, its HelixVPN instantiation, and its evidence shape. The `Tests:` columns in [06] and [07] are projections of §2 below.

Composition (per §11.4.169): §11.4.4(b) four-layer (pre-build gate + post-build

- runtime-on-clean-target + paired §1.1 mutation) applies to *every* item closure; §11.4.50 determinism (N=3 normal / N=10 cycle-validation, identical evidence hashes) applies to every PASS; §11.4.132 risk-ordering governs run order; §11.4.149 records each run in the item's `test_diary`; §11.4.135 registers a permanent regression guard for every closed defect.

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2. The closed test-type taxonomy (canonical)

This is the **single source** for the abbreviations used project-wide — **16 bundled families / 18 enumerated codes** (the families below expand to the 18 `test_type` CHECK codes of the §6 coverage-ledger schema: `UI`/`UX` and `CHAL`/`HQA` each render as one family but distinct codes). `DDOS` and `SCALE` are MVP-`NOT_APPLICABLE` by design (single-node self-host) and re-arm in Phase 2 (HA / managed).

Abbrev	§11.4.169 type	HelixVPN floor expectation	Primary harness	MVP stat
`UNIT`	unit	pure logic: policy compiler, IPAM allocator, MapResponse delta diff, MASQUE framing, status FSM. Mocks allowed **only here** (§11.4.27).	`cargo test`, `go test`, `dart test`	required
`INT`	integration	real Postgres+Redis+edge booted on-demand via `containers` submodule (§11.4.76); **no mocks**.	testcontainers via `containers/pkg/boot`	required
`E2E`	end-to-end	real control plane drives the netns rig; `curl` reaches an authorized LAN host through the real tunnel.	netns + nftables + netem rig [04_P0 §3]	required
`FA`	full-automation	self-driving, re-runnable `-count=3`, zero human-in-loop (§11.4.98).	`make spike` / `make qa` one-shot	required
`SEC`	security	RLS cross-tenant denial, key-never-leaves-FFI, mTLS, kill-switch, DNS-leak, DPI-evasion, secret-leak audit.	netns + pcap + `security` submodule	required
`DDOS`	DDoS / load-flood	handshake-flood + volumetric flood; edge fail-static + rate-limit holds.	`iperf3` flood + custom WG-init flood	**NOT_APP single-no selfhost** (
`STRESS`	stress	sustained load (≥100 iters / ≥30 s), ≥10 parallel, boundary inputs (§11.4.85).	`stress_chaos.sh` `ab_stress_*`	required
`CHAOS`	chaos	mid-flight SIGKILL / Redis drop / iface flap / partial write; recovery	`stress_chaos.sh` `ab_chaos_*`	required

Abbrev	§11.4.169 type	HelixVPN floor expectation	Primary harness	MVP stat
		restores consistent state (§11.4.85).		
`CONC`	concurrency / atomicity	concurrent IPAM allocs / enrollments / policy edits; no lost update, no double-allocation.	Go `-race` + Postgres txn harness	required
`RACE`	race / deadlock	data-race + deadlock detection on the hot paths (reconcile loop, edge verdict swap).	`cargo +nightly` loom + Go `-race` + tsan	required
`MEM`	memory	iOS NEPacketTunnelProvider RSS under the ~15 MB historical ceiling, ≥30% headroom; no leak over 24 h soak.	Instruments / `/proc` / `dumpsys meminfo`	required (critical)
`BENCH`	benchmarking	throughput, CPU-per-Gbps, handshakes/sec, p99 framing latency.	`bench.sh` [04_P0 §8] + `criterion`	required
`PERF`	performance	latency p50/p99 vs the SLO budget (§7.2); Go GC-tail watch on hot paths.	`bench.sh` + histogram metrics	required
`SCALE`	scaling	N simulated agents holding streams; convergence + bounded memory / 24 h.	agent-fuzzer + coordinator soak	partial (SL full Phase
`UI` / `UX`	UI / UX	Flutter widget + golden tests; flow walkthrough (enroll→connect→reach) with vision verdict.	`flutter test` + `panoptic` drive	required
`REC`	recorded-evidence	window-scoped MP4 (§11.4.154/.155) + media-validation pipeline verdict (§11.4.163).	`panoptic` capture → `vision_engine`	required (visible ite
`CHAL` / `HQA`	Challenge / HelixQA	a `challenges` / `helix_qa` bank entry scoring PASS only on captured evidence (§11.4.27/.107).	`challenges` + `helix_qa` submodules	required (

> **Why `DDOS/SCALE` defer.** [04_P1 §11] scopes the MVP to a single rootless host with no public multi-tenant surface; a DDoS suite at MVP would test a threat the deployment topology does not yet present (§11.4.3 topology-SKIP). The seam is built now (rate-limit token buckets [04 §T9],

> stateless fail-static edge [01 I3]) and the suite is authored as a **parked Phase-2 bank** so it re-arms mechanically when HA lands, never as an omission.

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3. The captured-evidence model (§11.4.5 / .69 / .107)

3.1 Evidence taxonomy mapped to §11.4.69 sink-side classes

Every HelixVPN feature maps to one §11.4.69 sink-side evidence class; a PASS cites an artifact whose *shape* the class fixes. This table is the contract `vision_engine` and the `challenges` banks score against.

Feature class	§11.4.69 class	Captured-evidence shape (the artifact `ab_pass_with_evidence` cites)
tunnel carries traffic	`network_throughput`	`iperf3 -J` JSON with `sum_received.bits_per_second > 0`; pcap with ≥1 decrypted inner packet
authorized reachability	`network_connectivity`	`curl` HTTP 200 body hash from the overlay netns + pcap SYN→SYN-ACK
default-deny	`network_connectivity` (negative)	pcap showing SYN out, **zero** SYN-ACK; edge `nft`/eBPF verdict-map counter `drop++`
kill-switch / DNS-leak	`wifi_link` / negative	host-side pcap during forced drop: **zero** non-loopback packets, **zero** :53 egress
transport escalation	`network_connectivity`	`StatusReport.transport=="masque-h3"` + tshark `http3` classification, no WG signature
policy reconcile latency	counter delta	`helix_reconcile_seconds` p99 histogram bucket < 1 s
revoke latency	counter delta	revoke-event-ts → edge-peer-removed-ts timing CSV, p99 < 1 s
key-never-leaves	negative	FFI boundary scan + log scan: zero private-key bytes cross FFI / hit any log
iOS NE memory	`MEM`	

Feature class	**§11.4.69 class**	**Captured-evidence shape (the artifact `ab_pass_with_evidence` cites)**
		Instruments `.trace` / `footprint` RSS series, peak < ceiling, ≥30% headroom
UI connect flow	`REC`	window-scoped MP4 (§11.4.154) + `vision_engine` OCR verdict on the ConnectButton state

3.2 §11.4.107 liveness battery (the no-frozen-frame rules, applied)

The §11.4.107 mandate — *a single captured frame/sample is not proof* — instantiates for HelixVPN as:

1. **Liveness over a window**, not a point. Throughput is sampled over ≥10 s; a one-shot `ping` reply is a pre-filter, never the proof (§11.4.107(1)).
2. **Independent counter-advance.** Goodput (`iperf3`) **and** an independent kernel WG counter (`wg show <if> transfer` rx/tx bytes advancing) must *both* move — a flat WG counter with moving `iperf3` means a decoy/loopback path (§11.4.107(2)).
3. **Not-stale cross-check.** After a transport escalation, the post-escalation pcap's first inner packet must not be a replay of the pre-escalation flow's last packet (§11.4.107(4)).
4. **Loading is a distinct state.** A handshake in progress is `Connecting`, not `Connected`; the FSM must reach `Connected` before liveness is judged (§11.4.107(3)); timeout+reachable ⇒ FAIL, timeout+unreachable ⇒ SKIP.
5. **Metamorphic relations** where there is no golden source: same LAN host reached over WG-UDP vs MASQUE must return the **same body hash**; a paused tunnel stops the counter; 2× clients ⇒ ~2× aggregate goodput (§11.4.107(8)).

3.3 Self-validated analyzers (§11.4.107(10)) — defeating B5

Every analyzer (pcap classifier, leak detector, OCR verdict, memory parser) ships with a **golden-good + golden-bad fixture pair** and is wired into the meta-test sweep. An analyzer that passes its golden-bad fixture is itself the bluff and FAILs `CM-ANALYZER-SELF-VALIDATED`.

```
// tests/analyzers/leak_detector_selfvalidation.rs
        (§11.4.107(10))
#[test] fn golden_good_clean_drop_passes() {
    let v = LeakDetector::new().scan_pcap("fixtures/
        killswitch_clean.pcap");
```

```rust
    assert_eq!(v.verdict, Verdict::Pass);          // a truly-
        clean drop PASSes
}
#[test] fn golden_bad_leaked_dns_fails() {
    let v = LeakDetector::new().scan_pcap("fixtures/
        killswitch_leaked_dns.pcap");
    assert_eq!(v.verdict,
        Verdict::Fail);          // a seeded :53 leak MUST FAIL
    assert!(v.findings.iter().any(|f| f.proto == "dns" && f.dport
        == 53));
}
```

The paired §1.1 mutation for this gate strips the `dport == 53` clause from the detector and asserts `golden_bad_leaked_dns_fails` now *passes the leak* → meta-test FAILs → the mutation is caught. This is the mechanical defeat of B5.

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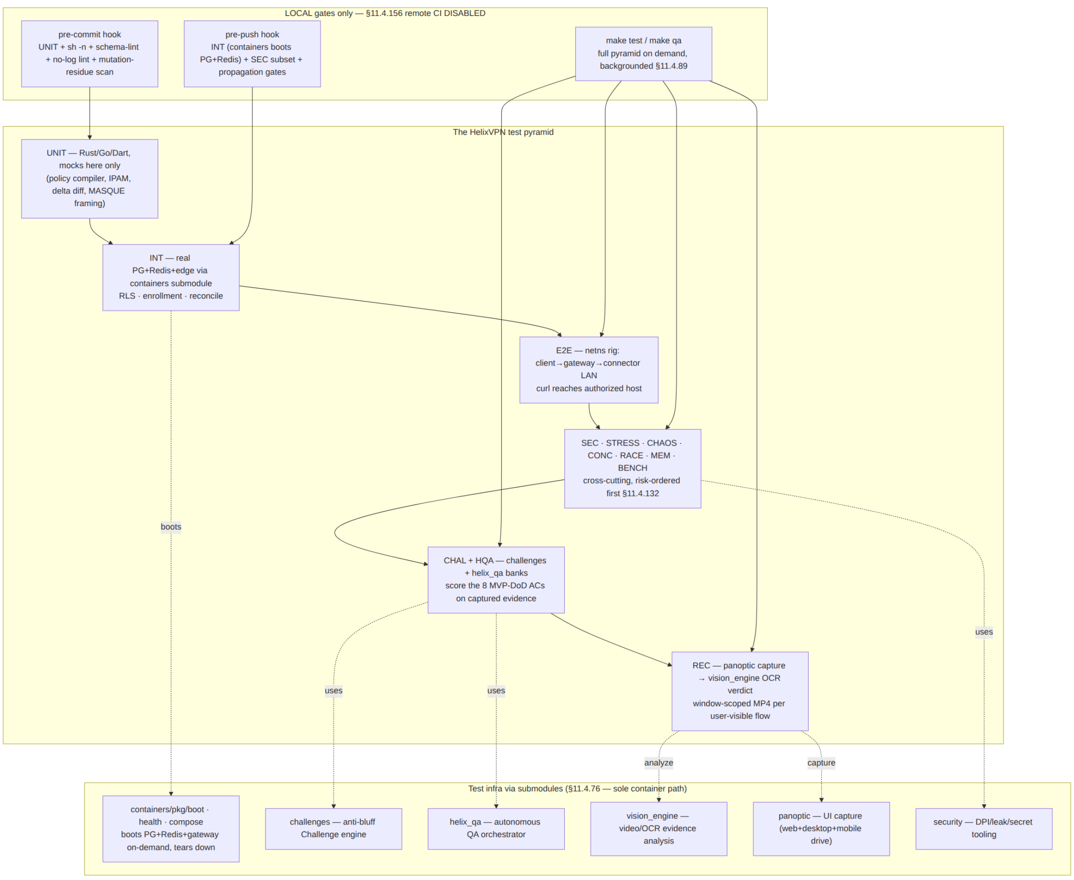
4. The test pyramid + ecosystem topology (§11.4.156 — local, no remote CI)



§11.4.156 binding. There is **no active `.github/workflows/*.yml` or `.gitlab-ci.yml`** in any HelixVPN repo. The CI workflow file is preserved at `.github/workflows/qa.yml.disabled-local-only` (a name no provider triggers, §11.4.75 Layer-5). Enforcement is the **local** git-hook ritual (§9) + the §11.4.40 pre-tag sweep run by the operator. Every gate in the pyramid is a Makefile target that runs on the developer's host; `containers` boots infra on demand so no shared runner is required.

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5. Per-type strategy, harness, evidence, gate, skeletons

5.1 UNIT

Covers (HelixVPN). Pure deterministic logic where a bug is a logic bug, not an integration bug: the policy compiler (ACL → AllowedIPs + verdict-map, default-deny correctness), the IPAM allocator (overlay `/48` ULA carve, collision-free allocation, D4 [SYNTHESIS]), the coordinator delta diff (snapshot→minimal-delta correctness), MASQUE/HTTP-Datagram framing encode/decode (RFC 9221 [research-masque]), the client status FSM transitions, kill-switch firewall-rule synthesis.

Harness. `cargo test` (Rust core/edge), `go test` (control plane), `dart test` (UI logic). Property tests via `proptest` (Rust) and `gopter` (Go) for the compiler and allocator — these are exactly the components where exhaustive hand-written cases miss the boundary that bites. **Mocks are permitted only at this layer** (§11.4.27).

Evidence. Test output + coverage report; for the compiler a *golden-table* fixture (`acl.yaml` → expected `AllowedIPs` set) committed as a regression guard (§11.4.135).

Gate. Pre-commit hook (§9). Coverage floor ratchets 70→85→95% over phases (§11.4.50 feature-coverage matrix).

```
// helix-core/crates/helix-policy/src/compile.rs  (interface under test)
pub trait PolicyCompiler {
    /// Compile a tenant ACL into per-peer AllowedIPs + an edge verdict map.
    /// MUST be default-deny: an empty/parse-failed ACL yields ZERO allowed routes.
    fn compile(&self, acl: &Acl, peers: &[PeerId]) ->
        Result<CompiledPolicy, CompileError>;
}

// helix-core/crates/helix-policy/tests/default_deny.rs
#[test] fn empty_acl_grants_nothing() {                        // the security floor
    let c = TailscaleAclCompiler::default();
```

```
    let cp = c.compile(&Acl::empty(), &[peer("alice"),
        peer("bob")]).unwrap();
    for p in cp.peers() { assert!(cp.allowed_ips(p).is_empty(),
        "fail-closed violated"); }
}
#[test] fn parse_error_is_fail_closed_not_fail_open() {       //
        B-class: never fail-open
    let c = TailscaleAclCompiler::default();
    assert!(matches!(c.compile(&Acl::malformed(), &[]),
        Err(CompileError::Parse(_))));
    // and the caller MUST treat Err as "no routes", proven in the
        INT layer (§5.2)
}
```

5.2 INT — integration (containers submodule infra)

Covers. Every seam that touches real infrastructure: Postgres RLS multi-tenant isolation (a tenant-B query under tenant-A's RLS context returns zero rows), device enrollment end-to-end (OIDC/anon token → device-generated WG keypair, private key never persisted server-side → short-lived mTLS cert issued), event backbone (Redis Streams publish → coordinator consume → delta emitted), the coordinator `WatchNetworkMap` server-stream (snapshot + deltas, peers already policy-filtered), no-log schema invariant against the live DB.

Harness — the `containers` submodule is the ONLY path (§11.4.76). Integration tests boot Postgres + Redis (+ edge where needed) **on-demand** via `containers/pkg/boot`, run against the *real* services, and tear down — no `docker`/`podman` CLI outside `pkg/boot`/`pkg/compose`/`pkg/health`, no mocks (§11.4.27). This is the on-demand-infra invariant: `make test` boots infra itself, the developer never runs `podman` by hand.

```
// helix-go/internal/store/rls_integration_test.go
//go:build integration
package store

import (
    "context"; "testing"
    "digital.vasic.containers/pkg/boot"     // §11.4.76 sole
        container seam
    "digital.vasic.containers/pkg/health"
)

func TestRLSCrossTenantDenial(t *testing.T) {
    ctx := context.Background()
    infra, err := boot.Up(ctx, boot.Spec{Postgres: true, Redis:
        true, Rootless: true}) // §11.4.161
    if err != nil { t.Fatalf("boot infra: %v", err) }
    t.Cleanup(func() { _ = infra.Down(ctx) })            //
        §11.4.14 quiescence
```

```go
    if err := health.WaitReady(ctx, infra, health.Postgres); err !
        = nil { t.Fatal(err) }

    db := mustConnect(t, infra.PostgresDSN())
    seedTenant(t, db, "tenant-A", "device-a1")
    seedTenant(t, db, "tenant-B", "device-b1")

    // Under tenant-A's RLS role, tenant-B's devices MUST be
        invisible (real DB, no mock).
    rows := queryAs(t, db, "tenant-A", `SELECT id FROM devices`)
    requireContains(t, rows, "device-a1")
    requireNotContains(t, rows, "device-b1")            //
        captured: rowset = evidence (§11.4.5)
    // paired §1.1 mutation: disable the RLS policy → this test
        MUST FAIL (gate CM-RLS-ENFORCED)
}
```

Evidence. The captured rowset / stream transcript; for the no-log invariant the `schemalint` report. **Gate.** Pre-push hook + `make test`.

```sql
-- the no-log invariant the INT + schema-lint layers enforce (§04,
        [02 §2.4])
-- CI schema-lint FAILS the build if ANY durable connection/
        traffic/packet table appears.
-- Allowed: aggregate counters only.
CREATE TABLE traffic_counters (              -- ALLOWED:
        aggregate, no per-flow rows
    tenant_id    uuid NOT NULL,
    window_start timestamptz NOT NULL,
    rx_bytes     bigint NOT NULL,             -- SUM over window,
        not per-connection
    tx_bytes     bigint NOT NULL,
    PRIMARY KEY (tenant_id, window_start)
);
-- FORBIDDEN (schemalint paired §1.1 mutation seeds this → gate
        MUST FAIL):
-- CREATE TABLE connections (src_ip inet, dst_ip inet, started_at
        timestamptz, ...);
```

5.3 E2E — end-to-end (netns rig)

Covers. The whole reachability slice on one box: a *real* control plane issues a NetworkMap, a *real* `helix-core` client builds the tunnel, traffic crosses `client → gateway → connector LAN`, and `curl http://10.10.0.20/` returns the hello page. Plus the *negative* E2E (default-deny: a SYN out, zero SYN-ACK) and the DPI-block E2E (plain WG blocked → MASQUE escalation → reachability preserved). The rig is the reproducible substrate for AC2/AC3/AC4 (§7.2) and survives Phase-0→1.

Harness — Linux netns + nftables-DPI + tc-netem [04_P0 §3, 06 §HVPN-P0-022]. Three processes (`client`, `gateway`, `connector`) in namespaces on one host; an `nft` table simulates a DPI UDP block; `tc netem` injects loss/delay for resilience.

```
# rig/netns_up.sh — connector-side "private LAN" simulated with a netns [06 HVPN-P0-023]
set -euo pipefail
ip netns add lanA
ip link add veth-host type veth peer name veth-lanA
ip link set veth-lanA netns lanA
ip -n lanA addr add 10.10.0.20/24 dev veth-lanA && ip -n lanA link set veth-lanA up
ip netns exec lanA python3 -m http.server 80 --bind 10.10.0.20 &   # the hello service (sink)

# rig/dpi_block.sh — DPI/censorship sim for the AC4 escalation E2E
nft add table inet dpi
nft add chain inet dpi fwd '{ type filter hook forward priority 0; }'
nft add rule  inet dpi fwd udp dport 51820 drop      # kill plain WireGuard
nft add rule  inet dpi fwd udp dport 443  accept     # allow MASQUE/H3

# rig/impair.sh — loss/jitter for the resilience E2E (§11.4.85 adjacency)
tc qdisc add dev "$IFACE" root netem loss 5% delay 40ms 10ms

# rig/test_reach.sh — the E2E assertion (positive + negative), §11.4.50 N=3 deterministic
run_reach() {                                        # $1 = expected: pass|deny
  pcap="qa-results/e2e/$(date +%s)_reach.pcap"
  ip netns exec client tcpdump -i overlay0 -w "$pcap" &  TPID=$!
  body=$(ip netns exec client curl -s --max-time 5 http://10.10.0.20/ || true)
  kill "$TPID" 2>/dev/null
  if [ "$1" = pass ]; then
    [ -n "$body" ] || ab_fail "B1: no body — config-only PASS forbidden"
    grep -q "SYN-ACK"  <(tshark -r "$pcap" -Y 'tcp.flags.syn==1 && tcp.flags.ack==1') \
      && ab_pass_with_evidence "authorized reach" "$pcap"
  else
    [ -z "$body" ] || ab_fail "default-deny breached: got a body"
    tshark -r "$pcap" -Y 'tcp.flags.syn==1 && tcp.flags.ack==1' | grep -q . \
      && ab_fail "deny breached: SYN-ACK present" \
      || ab_pass_with_evidence "default-deny (no SYN-ACK)" "$pcap"
  fi
}
```

Evidence. pcap + `curl` body hash + `iperf3 -J` CSV (≥80% bare-link for G1). **Gate.** `make test` E2E stage; AC2/AC3/AC4 release gates (§7.2). Cleanup is non-negotiable — `trap 'rig/netns_down.sh' EXIT` leaves the host quiescent (§11.4.14).

5.4 FA — full-automation

Covers. The §11.4.98 invariant: **every** E2E / integration / Challenge must be re-runnable end-to-end with **zero human intervention** after startup. For HelixVPN this means the whole MVP-DoD demo (`helixvpnctl init` → enroll connector+client → reach → deny → escalate → revoke) runs as one command, N=3 consecutively, with self-cleaning state and identical PASS each time (§11.4.50). The single forbidden human touch-point is credential bootstrap *outside* execution (a `.env` token, §11.4.10).

Harness. `make spike` (Phase 0) and `make qa` (Phase 1) are the one-shot drivers; the latter dispatches `helix_qa`'s autonomous session (§5.6). Determinism is mechanical: `ab_run_n_times <name> 3 <fn>` loops, captures an evidence-hash per iteration, and FAILs if any hash or exit code diverges — there is no "first-pass-was-a-flake" path (§11.4.50).

```
# the §11.4.50 determinism wrapper every FA suite uses
ab_run_n_times() {                           # $1 name  $2 N  $3 fn
  local name="$1" n="$2" fn="$3" h prev=""
  for i in $(seq 1 "$n"); do
    "$fn" "$i"; h=$(sha256sum "qa-results/$name/run_$i/
        evidence.json" | cut -d' ' -f1)
    [ -z "$prev" ] || [ "$h" = "$prev" ] || ab_fail "non-
        deterministic: run $i hash diverged"
    prev="$h"
  done
  ab_pass_with_evidence "$name x$n deterministic" "qa-results/
        $name/"
}
```

5.5 CHAL — Challenges (challenges submodule banks)

Covers. The eight MVP Definition-of-Done acceptance criteria (AC1–AC9, §7.2) are authored as **executable Challenges** in the `challenges` submodule — the constitution's anti-bluff Challenge layer (§11.4.27(B)/.5/.69). A Challenge differs from a test: it scores PASS **only** on captured evidence matching the §3.1 shape for that feature class, and is graded by the `challenges` engine, not by the test's own exit code. This is the structural defence against B1–B3: the Challenge engine re-reads the pcap/counter/recording, it does not trust the harness's claim.

Harness. challenges submodule (digital.vasic.challenges, Go) consumed via replace in dev / pinned SHA in release [05 §6.1]. A bank file maps each AC to its on-rig driver + the evidence-class assertion.

```yaml
# challenges/banks/helixvpn_mvp_dod.yaml  (illustrative — one entry per AC)
bank: helixvpn-mvp-dod
challenges:
  - id: HVPN-CHAL-AC2-authorized-reach
    feature_class: network_connectivity       # §11.4.69
    driver: rig/test_reach.sh pass
    evidence:
      kind: pcap
      assert: "frame.tcp.flags.syn==1 && frame.tcp.flags.ack==1 from 10.10.0.20"
      and_body_hash_nonempty: true
    score: pass_only_if_evidence_matches      # never trust exit code alone
  - id: HVPN-CHAL-AC7-killswitch-no-leak
    feature_class: killswitch_negative
    driver: rig/killswitch_drop.sh
    evidence:
      kind: pcap
      assert_zero: "ip && not loopback"       # zero plaintext egress
      assert_zero_dns: "udp.port==53"         # zero DNS leak
    self_validated: true                       # golden-bad seeded-leak pcap MUST score FAIL (§3.3)
```

Evidence + Gate. The Challenge engine's per-AC verdict JSON under docs/qa/<run-id>/ (§11.4.83); make qa is the gate; a release tag is blocked while any DoD Challenge is non-PASS.

5.6 HQA — HelixQA autonomous sessions

Covers. helix_qa is the **autonomous QA orchestrator** (§11.4.27/.107/.158): it drives the full MVP-DoD acceptance set end-to-end with no human, detects and respawns crashed sub-runs (§11.4.147), generates tickets for findings into the workable-items DB, and emits the real-time sync channel (§11.4.116 JSONL event stream + atomic status snapshot) that the conductor tails. It is the §11.4.165 independent-verification agent for the runtime layer — structurally separate from the code author.

Harness. go run ./submodules/helix_qa/cmd/orchestrator --suite mvp-dod ([05 Makefile qa:]). It consumes the challenges banks (§5.5), boots infra via containers (§5.2), drives panoptic for UI flows (§5.14), and feeds frames to vision_engine (§5.14). Each verdict event carries its evidence path (§11.4.116 — a PASS with no evidence path is a contradiction → treated as FAIL).

```
// the §11.4.116 sync channel helix_qa emits (qa-results/helix_qa/
events.jsonl, append-only)
{"ts":"2026-…","ev":"session_start","suite":"mvp-
dod","build":"<artifact-md5>"}
{"ts":"…","ev":"challenge_start","id":"HVPN-CHAL-AC2-authorized-
reach"}
{"ts":"…","ev":"evidence","id":"HVPN-CHAL-AC2-…","path":"qa-
results/.../reach.pcap"}
{"ts":"…","ev":"verdict","id":"HVPN-CHAL-
AC2-…","result":"PASS","evidence":"qa-results/.../reach.pcap"}
```

Evidence + Gate. The events stream + per-AC verdict; `make qa` exit status gates the release. HelixQA's own meta-test plants a known-broken AC and asserts the session reports FAIL (§11.4.32 self-meta-test).

5.7 SEC — security

Covers. The security invariants from [04 §13], each a falsifiable captured-evidence row:

Invariant	Captured-evidence proof
S1 default-deny	negative E2E (§5.3): SYN out, zero SYN-ACK, edge `drop++`
S2 key-never-leaves	FFI boundary scan + log scan: **zero** private-key bytes cross FFI or hit any log [04 §13]
S3 mTLS device cert	handshake pcap shows client cert; expired/ revoked cert ⇒ stream refused
S4 kill-switch	host pcap during forced tunnel drop: zero non-loopback egress [04, §11.4.68]
S5 DNS-leak	host pcap during drop + reconnect: zero :53 to a non-tunnel resolver
S6 DPI-evasion	tshark classifies MASQUE flow as HTTP/3, **no** WG signature, SNI = decoy host
S7 wire-fingerprint	passive observer cannot distinguish HelixVPN MASQUE from generic H3 (entropy + timing)
S8 secret-leak audit	`git ls-files | xargs grep -l <secret>` + `git log -S<secret>` empty (§11.4.10/.10.A)
S9 RLS cross-tenant	§5.2 INT rowset isolation

Harness. netns + pcap for S1/S3–S7; the `security` submodule for the DPI/leak/ secret tooling [05 §6 row]; FFI scan tool for S2. Each analyzer is self-validated (§3.3). **Gate.** Pre-push hook (secret audit + RLS subset) + `make qa` SEC stage. The secret-leak audit is also a §11.4.10.A pre-store audit run before any credential lands.

```
# rig/killswitch_drop.sh — S4+S5, the anti-bluff kill-switch test
        (defeats B2)
set -euo pipefail
pcap="qa-results/sec/$(date +%s)_killswitch.pcap"
ip netns exec client tcpdump -i any -w "$pcap" & TPID=$!
ip netns exec client curl -s --max-time 30 http://10.10.0.20/ >/
        dev/null &  # traffic in flight
sleep 2
rig/force_tunnel_drop.sh                       # core FSM ->
        Blocked, firewall seals
sleep 5
ip netns exec client nslookup example.com 2>/dev/null || true   #
        try to leak a DNS query
kill "$TPID" 2>/dev/null
# PASS only if, after the drop, ZERO non-loopback packets AND ZERO
        :53 left the host:
LEAK=$(tshark -r "$pcap" -Y 'frame.time_relative>2 && ip && not
        (ip.addr==127.0.0.1)' | wc -l)
DNS=$(tshark  -r "$pcap" -Y 'frame.time_relative>2 &&
        udp.port==53' | wc -l)
[ "$LEAK" -eq 0 ] && [ "$DNS" -eq 0 ] \
  && ab_pass_with_evidence "kill-switch sealed, no DNS leak"
        "$pcap" \
  || ab_fail "LEAK=$LEAK DNS=$DNS — plaintext/DNS escaped the
        seal"
```

5.8 DDOS — denial-of-service / load-flood

Covers (Phase 2; seam built MVP). Two threats: (a) **handshake flood** — a torrent of WG/MASQUE init packets must not exhaust the gateway (rate-limited via Redis token buckets [04 §T9]); (b) **volumetric flood** — the stateless edge must fail-static (drop, never crash; rootless auto-restart [01 I3]). At MVP this is `NOT_APPLICABLE: single-node-selfhost` (§11.4.6) and **authored as a parked bank** that re-arms in Phase 2 when the public multi-tenant surface exists.

Harness (parked). `iperf3` N-client flood + a custom Noise-IK-init flood generator; assert p99 handshake latency for a *legitimate* client stays bounded while the flood runs, and the edge process never OOM/crashes (counter `wg_init_dropped` rises, `helix_edge_up` stays 1). **Evidence.** legit-client latency CSV + edge liveness counter during flood. **Gate.** Phase-2 release gate; MVP records the SKIP reason in the coverage ledger (§6).

5.9 STRESS + CHAOS

Covers. Resilience of every fix (§11.4.85 mandate — happy-path-only is a bluff at the resilience layer). **STRESS:** sustained reconcile churn (≥100 policy edits / ≥30 s), ≥10 concurrent enrollments, boundary inputs (empty ACL, max-peer map, off-by-one overlay range). **CHAOS:** mid-transfer SIGKILL of the edge (tunnel recovers, no plaintext leak during the gap), Redis drop mid-reconcile (coordinator degrades gracefully, no lost delta), interface flap (client roams, tunnel re-establishes < 3 s), partial-write of the network-map snapshot (recovery restores a consistent map, never a torn one), connector process kill (peer drops from every map within the convergence budget).

Harness. `stress_chaos.sh` exposing `ab_stress_run`, `ab_stress_concurrent`, `ab_chaos_kill_pid_during`, `ab_chaos_drop_network_during`, `ab_chaos_iface_flap_during` — each composing with `ab_pass_with_evidence` and a `trap '...' EXIT` cleanup (§11.4.14 — corrupt-restore / process-restart mandatory).

```
# CHAOS: kill the edge mid-transfer, assert recovery AND no leak
        during the gap (§11.4.85)
ab_chaos_kill_pid_during "$(edge_pid)" 'ip netns exec client
        iperf3 -t 20 -c 10.10.0.20' \
  --on-kill 'sleep 3' \
  --assert 'tunnel_reestablished_within 3s &&
        killswitch_sealed_during_gap' \
  --evidence "qa-results/chaos/edge_kill_$(date +%s)/"
# the assertion re-reads the gap pcap: zero plaintext egress while
        the edge was dead (composes S4)
```

Evidence. `latency.json` (p50/p95/p99), `recovery_trace.log`, `state_delta_snapshot.json`, gap-pcap. **Gate.** `make test` STRESS/CHAOS stage; every closed defect registers a permanent CHAOS guard (§11.4.135).

5.10 CONC — concurrency / atomicity

Covers. The places where two callers race over shared state: concurrent IPAM allocation (no two devices get the same overlay IP — a Postgres `SELECT … FOR UPDATE` / unique-constraint contract), concurrent enrollment of the same device, concurrent policy edits to the same tenant (last-writer-wins with a version guard, no lost update), concurrent `WatchNetworkMap` streams receiving deltas while a new peer enrolls (each stream converges to the same final map). The mandate: any method reachable from ≥2 goroutines/threads is safe for rapid consecutive calls (project principle 3).

Harness. Go test spawning N goroutines hammering the allocator against real Postgres (via `containers`), asserting **exactly** N distinct IPs and zero constraint violations; a property test that the final map is invariant to delta arrival order.

```
func TestIPAMNoDoubleAllocationUnderConcurrency(t *testing.T)
        { // CONC + atomicity
    infra := mustBootPG(t)                          // containers
        submodule (§5.2)
    alloc := NewIPAMAllocator(infra.PostgresDSN(),
        mustParseULA("fd7a:115c:a1e0::/48"))
    const N = 64
```

```
    got := make([]netip.Addr, N); var wg sync.WaitGroup
    for i := 0; i < N; i++ { wg.Add(1); go func(i int){ defer
        wg.Done();
        a, err := alloc.Allocate(ctx, deviceID(i));
        requireNoErr(t, err); got[i] = a }(i) }
    wg.Wait()
    requireAllDistinct(t, got)                    // captured: the
        N-set is the evidence
    // paired §1.1 mutation: drop the FOR UPDATE lock → a
        collision appears → gate FAILs
}
```

Evidence. the allocated-set + Postgres constraint-violation count (must be 0). **Gate.** `make test` CONC stage; run with Go `-race` (overlaps §5.11).

5.11 RACE — race-condition / deadlock

Covers. Data races and deadlocks on the hot paths: the client reconcile loop (network-map apply under a concurrent status read — no torn read of `AllowedIPs`), the edge verdict-map hot-swap (a policy update swaps the verdict map while packets are being classified — no use-after-free, no half-applied map), the Rust core's `tick()` keepalive/rekey timer under concurrent send/recv, the Go coordinator's in-memory topology graph under concurrent stream fan-out.

Harness. Rust: `cargo +nightly test --features loom` (exhaustive interleaving of the reconcile/verdict-swap critical sections) + ThreadSanitizer on the FFI boundary. Go: `go test -race` on the coordinator + edge. The §11.4 principle 2 (no blocking op inside a held lock) is asserted by a loom test that the verdict-swap never holds the classify lock across an allocation.

```
// helix-edge/tests/verdict_swap_loom.rs — no torn read, no
        deadlock under interleaving
#[test] fn verdict_swap_is_atomic_under_loom() {
    loom::model(|| {
        let vm =
        Arc::new(ArcSwap::from_pointee(VerdictMap::empty()));
        let (vm1, vm2) = (vm.clone(), vm.clone());
        let t = loom::thread::spawn(move || {
        vm1.store(Arc::new(VerdictMap::allow("10.10.0.0/24"))); });
        let v =
        vm2.load();                    // classifier reads
        concurrently
        assert!(v.is_consistent());           // never a half-
        applied map (no torn read)
        t.join().unwrap();
    });
}
```

Evidence. loom/tsan/`-race` clean report (zero findings). **Gate.** `make test` RACE stage; a `-race` finding is a release blocker (B-class latent bug per §11.4.145 angle 3).

5.12 MEM — memory (iOS NE soak)

Covers — the make-or-break platform constraint (Phase-0 gate G3 [04_P0]). iOS `NEPacketTunnelProvider` runs under a hard memory ceiling (~15 MB historical [SYNTHESIS §5]); the entire decision to write `helix-core` in **Rust not Go** (D2 [SYNTHESIS §3]) rests on staying under it with ≥30% headroom. MEM also covers: no RSS growth over a 24 h soak (no leak in the reconcile loop or the QUIC buffer pool), the coordinator's bounded memory at 10k streams (SLO4 §7.2), and Android `VpnService` RSS.

Harness. iOS: Instruments `Allocations/Leaks` + `footprint` against a device build, peak-RSS series captured as a `.trace`; a soak harness runs a sustained transfer for 24 h sampling `/proc/<pid>/status` (Linux core), `dumpsys meminfo` (Android), Instruments (iOS). The G3 gate: peak RSS < ceiling, headroom ≥ 30%, proven on a **real device** (a simulator does not reproduce the jetsam ceiling — §11.4.3: no device ⇒ honest `SKIP: hardware_not_present`, never a fake PASS).

```
// shims/apple/PacketTunnelProvider+MemoryProbe.swift  (G3
        evidence emitter)
func sampleFootprint() -> UInt64 {              // bytes; written
        to qa-results/mem/ios_rss.csv
    var info = task_vm_info_data_t()
    var count =
        mach_msg_type_number_t(MemoryLayout<task_vm_info>.size /
        MemoryLayout<integer_t>.size)
    let kr = withUnsafeMutablePointer(to: &info) {
        $0.withMemoryRebound(to: integer_t.self, capacity:
        Int(count)) {
            task_info(mach_task_self_,
        task_flavor_t(TASK_VM_INFO), $0, &count) } }
    return kr == KERN_SUCCESS ? info.phys_footprint : 0   //
        phys_footprint = the jetsam metric
}
```

Evidence. RSS-vs-time CSV + Instruments `.trace`; G3 verdict = peak < ceiling × 0.7. **Gate.** Phase-0 G3 (release-blocking make-or-break); MVP soak for SLO4.

5.13 BENCH + PERF + SCALE

Covers. BENCH: raw throughput (≥80% bare-link plain-UDP G1; ≥50% of plain over MASQUE G2 [04_P0]), CPU-per-Gbps for both candidate edge languages (the Go-vs-Rust G4 decision input, D5 [SYNTHESIS]), handshakes/sec, MASQUE framing latency. **PERF:** the four SLOs as p99 budgets (§7.2) — event→delta < 1 s, enroll→first-map < 2 s, revoke→enforcement < 1 s; Go GC-tail watch on the coordinator hot path. **SCALE:** N simulated agents holding `WatchNetworkMap` streams; convergence p99 < 1 s and bounded coordinator memory over a 24 h soak (SLO4).

Harness. `bench.sh` [04_P0 §8] driving the netns rig with `iperf3 -J` + per-iteration CSV, run N=3 deterministic (§11.4.50). `criterion` for Rust micro-benches (framing). A coordinator agent-fuzzer holds N streams and measures convergence + RSS.

```
# bench.sh — the G1/G2/G4 benchmark, deterministic over N=3 [04_P0
        §8, 06 HVPN-P0-NNN]
THRPUT=$(iperf3 -J -c 10.10.0.20 -P "$CLIENTS" | jq
        '.end.sum_received.bits_per_second')
GOODPUT_LOSS=$(with_netem 'loss 5% delay 40ms'
        iperf3_goodput)       # G2 / resilience
CPU_PER_GBPS=$(awk -v t="$THRPUT" -v c="$EDGE_CPU_SEC"
        'BEGIN{print c/(t/1e9)}')  # G4 input
printf '%s,%s,%s,%s,%s\n' "$EDGE_LANG" "$RUN" "$THRPUT"
        "$GOODPUT_LOSS" "$CPU_PER_GBPS" \
  >> qa-results/bench/edge_compare.csv      # the G4 decision CSV
        (rust vs go)
```

Evidence. the per-metric CSV with min/max/mean/p95 (§11.4.24 resource-stats discipline); the G4 CSV decides `gates.G4.outcome IN ('rust','go')`. **Gate.** Phase-0 G1/G2/G4; MVP SLO PERF/SCALE gates; a PERF regression > budget is a release blocker.

5.14 UI + UX + REC — recorded-evidence (vision_engine / panoptic)

Covers. The three Flutter flavors (Access / Connector / Console [04_UI]): **UI** = widget + golden tests on the design-system components (ConnectButton, StatusChip, ExitPicker, ShieldIndicator) across light+dark (§11.4.162 OpenDesign parity); **UX** = the real user journey (enroll → connect → reach → see the shield go green) driven through the app's own UI; **REC** = a **window-scoped MP4** (§11.4.154/.155 — app window only, never whole desktop) of that journey, fed to the media-validation pipeline (§11.4.163) for an OCR/vision verdict.

Harness. `flutter test` (widget + golden). `panoptic` (UI capture harness, Go [05 §6]) drives the web/desktop/mobile app and captures the window-scoped MP4; `vision_engine` (Go) runs OCR/state-verify on the frames — e.g. it reads that the ConnectButton text transitions `Connect → Connecting → Connected` and the ShieldIndicator turns green, then cross-checks against the *real* core status stream (a green shield while the core FSM is `Blocked` is B3 and FAILs). Filenames carry the §11.4.155 project prefix `helixvpn---connect-flow---<run-id>.mp4`; raw corpus is git-ignored, curated evidence committed under `docs/qa/<run-id>/` (§11.4.83).

```dart
// helix-ui/test/connect_button_golden_test.dart  (UI — light+dark, §11.4.162)
void main() {
  for (final brightness in [Brightness.light, Brightness.dark]) {
    testWidgets('ConnectButton golden — $brightness', (tester) async {
      await tester.pumpWidget(HelixTheme(brightness: brightness,
          child: ConnectButton(state: ConnState.connected)));
      await expectLater(find.byType(ConnectButton),
          matchesGoldenFile('goldens/connect_button_connected_$brightness.png'));
    });
  }
}
```

```
// the §11.4.163 media-validation verdict vision_engine emits for the REC artifact
{ "artifact": "docs/qa/<run-id>/helixvpn---connect-flow---<run-id>.mp4",
  "expected_patterns": ["Connect","Connecting","Connected","shield:green"],   // SPECIFY-phase (§11.4.159(J))
  "matched": ["Connect","Connecting","Connected","shield:green"],
  "cross_check": { "core_fsm_at_green_shield": "Connected" },                  // defeats B3
  "verdict": "PASS", "self_validated": true }                                 // golden-bad MP4 → FAIL (§3.3)
```

Evidence. golden PNGs + the window-scoped MP4 + the vision verdict JSON. **Gate.** `make test` UI stage + `make qa` REC stage; AC9 (three apps drive it). A user-visible item without a vision-verified REC is a §11.4.153 release blocker.

---

6. The coverage ledger (feature × test-type × evidence-state)

The coverage ledger is the §11.4.25/.52/.153 mechanical answer to "is every feature proven, by every warranted test type, with real evidence?" It is a **git-tracked SQLite table** (§11.4.93/.95) — a projection of the workable-items DB cross-joined with the test-type matrix — regenerated by `doc_processor` from the spec feature map (§11.4.153) on every QA run, and rendered to `docs/qa/coverage_ledger.md` (+HTML/PDF) by `docs_chain` (§11.4.106).

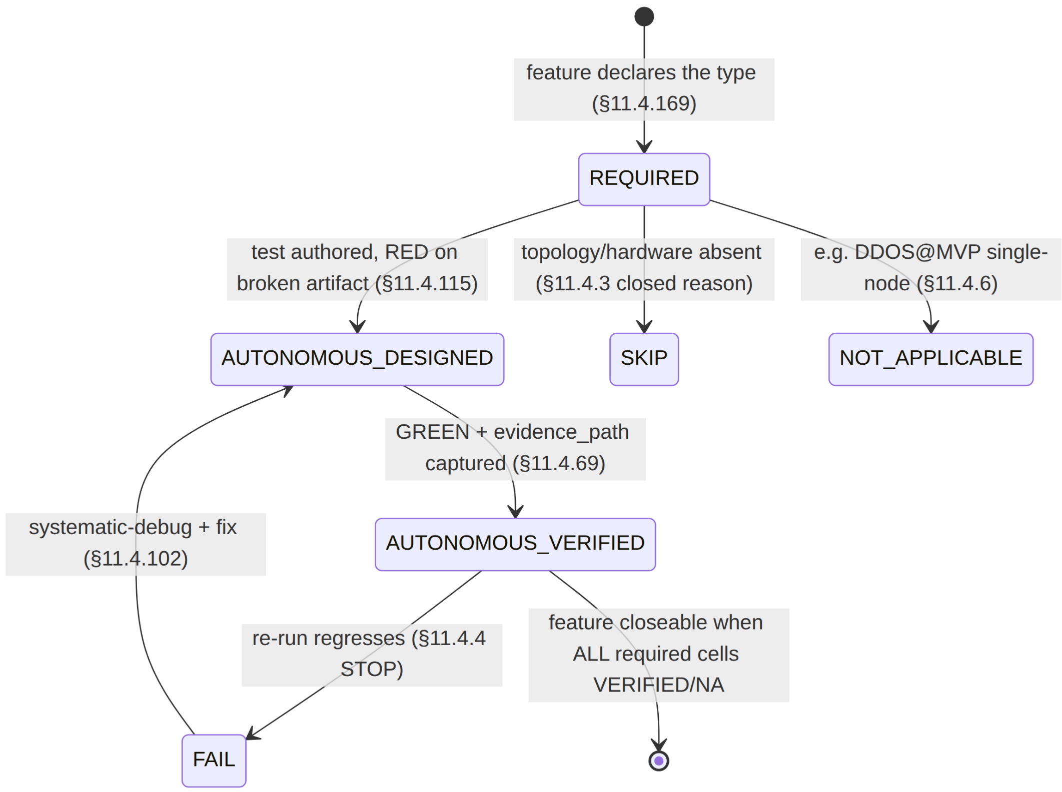
6.1 DDL

```sql
-- docs/.workable_items.db  (git-tracked, §11.4.95) — coverage
        ledger projection
CREATE TABLE IF NOT EXISTS features (
    feature_id   TEXT PRIMARY KEY,                 -- 'F-
        KILLSWITCH', 'F-AUTHZ-REACH'
    component    TEXT NOT NULL,                     -- 'helix-
        core' | 'helix-go' | 'helix-edge' | 'helix-ui'
    title        TEXT NOT NULL CHECK (length(title) >= 40),   --
        §11.4.91
    evidence_class TEXT NOT NULL,                   -- §11.4.69
        class (network_connectivity, ...)
    phase        TEXT NOT NULL CHECK (phase IN
        ('P0','P1','P2','P3'))
);
CREATE TABLE IF NOT EXISTS coverage_cells (
    feature_id   TEXT NOT NULL REFERENCES features(feature_id),
    test_type    TEXT NOT NULL,                     -- §2 abbrev:
        UNIT|INT|E2E|SEC|...
    state        TEXT NOT NULL CHECK (state IN

        ('REQUIRED','AUTONOMOUS_VERIFIED','AUTONOMOUS_DESIGNED',

        'OPERATOR_ATTENDED_ONLY','SKIP','NOT_APPLICABLE','FAIL','PENDING')),
    skip_reason  TEXT,                              -- closed set
        (§11.4.3) when state IN (SKIP,NOT_APPLICABLE)
    evidence_path TEXT,                             -- §11.4.69 —
        REQUIRED for *_VERIFIED states
    last_run_at  TEXT,
    diary_ref    TEXT,                              -- §11.4.149
        test_diary row id
    PRIMARY KEY (feature_id, test_type),
    -- §11.4.69: a VERIFIED cell MUST carry an evidence path
        (schema makes a bluff impossible)
    CHECK (state NOT IN ('AUTONOMOUS_VERIFIED') OR (evidence_path
        IS NOT NULL AND length(evidence_path)>0)),
    -- §11.4.3: a SKIP/NA cell MUST carry a closed-set reason
    CHECK (state NOT IN ('SKIP','NOT_APPLICABLE') OR (skip_reason
        IS NOT NULL))
);
CREATE VIEW v_coverage_gaps AS
  SELECT f.feature_id, c.test_type, c.state
  FROM features f JOIN coverage_cells c USING (feature_id)
  WHERE c.state IN ('PENDING','FAIL','OPERATOR_ATTENDED_ONLY');
        -- the release-blocker set
```

6.2 State machine for a cell



Release rule. A phase ships only when `v_coverage_gaps` is empty for that phase (every required cell is `AUTONOMOUS_VERIFIED` or `NOT_APPLICABLE` with a reason); `OPERATOR_ATTENDED_ONLY` is a release blocker until promoted with a tracked migration item (§11.4.52). The ledger is the §11.4.40 full-suite-retest scoreboard.

6.3 Worked rows (illustrative)

feature_id	UNIT	INT	E2E	SEC	STRESS	CHAOS	CONC	RACE	M
F-AUTHZ-REACH	✓	✓	✓	✓	✓	✓	—	—	—
F-DEFAULT-DENY	✓	✓	✓	✓	—	✓	—	✓	—
F-KILLSWITCH	✓	—	✓	✓	—	✓	—	✓	—
F-TRANSPORT-ESCALATE	✓	✓	✓	✓	✓	✓	—	✓	—
F-POLICY-RECONCILE	✓	✓	✓	—	✓	✓	✓	✓	—
F-REVOKE	✓	✓	✓	✓	—	✓	✓	—	—
	—	—	—	—	—	—	—	—	✓

feature_id	UNIT	INT	E2E	SEC	STRESS	CHAOS	CONC	RACE	M
F-IOS-NE-MEM									
F-NO-LOG-SCHEMA	✓	✓	—	✓	—	—	—	—	—
F-DDOS-FLOOD	—	—	—	NA[1]	—	—	—	—	—

[1] `NOT_APPLICABLE: single-node-selfhost` at MVP; re-arms Phase 2 (§5.8).

---

7. Per-phase acceptance gates

7.1 Phase 0 — Spike exit gates (G1–G6) [04_P0, 06]

Gate	Question	Go/No-Go bar	Evidence	Test types
G1	plain-UDP WG client→gw→connector LAN?	iperf3 ≥ **80%** bare-link; `curl 10.10.0.20:80` OK	pcap + iperf3 CSV	E2E, BENCH, SC, FA, CH
G2	same core over MASQUE/QUIC through a DPI UDP block?	≥ **50%** of plain-UDP; survives 5% loss > UoT strawman; tshark classifies HTTP/3, no WG sig	pcap + tshark + CSV	E2E, SEC, BENCH, FA
G3	iOS NE Rust core under memory ceiling? **(make-or-break)**	peak RSS < ceiling, **≥30%** headroom on a real device	Instruments `.trace` + RSS CSV	MEM
G4	Go-vs-Rust edge benchmark?	decision recorded from CPU-per-Gbps/ p99/churn/ mem CSV (N=3)	`edge_compare.csv`	BENCH, PERF, RACE

Gate	Question	Go/No-Go bar	Evidence	Test types
G5	flutter_rust_bridge FFI drives core from Dart?	Dart calls `connect()`, receives status stream	FFI round-trip log + UI recording	UNIT, UI, FA
G6	push-based reconcile from a static map?	a map edit reconciles the slice with no restart	reconcile event log	INT, E2E, FA

Each gate is a row in the `gates` table ([06 §0.3]); `outcome` flips to `pass`/`fail`/ `rust/go` only with an `evidence_path`. A gate that cannot clear inside the time box **is the finding** — escalate the decision (§11.4.66), never overrun silently.

7.2 Phase 1 — MVP Definition-of-Done (AC1–AC9 + SLO1–SLO4) [04_P1, 07]

The MVP ships only when **all nine ACs and all four SLOs** pass on a clean baseline (§11.4.40 retest + §11.4.108 runtime-signature on a fresh deploy). Each AC is a `challenges` bank entry (§5.5) scored by `helix_qa` (§5.6).

AC	Criterion	Captured evidence	Bound i
AC1	self-host from zero (`helixvpnctl init` + start)	recorded terminal, healthchecks green, `helix_up==1`	HVPN-P1
AC2	enroll connector+client; reach authorized LAN host	E2E netns capture + pcap (body hash)	F-AUTHZ REACH
AC3	DENY unauthorized host (default-deny)	negative E2E: SYN out, zero SYN-ACK	F-DEFAU DENY
AC4	auto-escalate to MASQUE when WG blocked, stays up	`StatusReport.transport=="masque-h3"` + tshark H3	F-TRANSP ESCALAT
AC5	policy edit reconfigures all devices < 1 s, no restart	`helix_reconcile_seconds` p99 < 1 s	F-POLICY RECONC
AC6	revoke device < 1 s	revoke→edge-enforcement timing CSV	F-REVOK
AC7	kill-switch + DNS-leak: no plaintext on drop	host pcap, zero leak / zero :53 (§5.7)	F-KILLSWI
AC8	no durable conn/traffic log (schema-lint green)	`schemalint` PASS + §1.1 mutation FAIL	F-NO-LO SCHEMA
AC9	three apps drive it all (Access+Connector+Console)	per-app window-scoped UX recordings + vision verdict	F-UI-FLO

SLO	Target	Test type	Item
SLO1 event→delta-on-wire	p99 < 1 s	PERF	HVPN-P1-073
SLO2 enroll→first NetworkMap	< 2 s	PERF	HVPN-P1-031
SLO3 revoke→edge enforcement	< 1 s	PERF	HVPN-P1-033
SLO4 coordinator memory @ 10k streams	bounded / 24 h soak	MEM, SCALE	HVPN-P1-074

7.3 Phase 2 / 3 — re-armed and new gates

Phase 2 **re-arms** `DDOS` + full `SCALE` (HA multi-region, stateless coordinators, Patroni PG, NATS JetStream) and adds: DAITA traffic-shaping evidence (maybenot), P2P NAT-traversal E2E (hole-punch success rate over a STUN/DERP rig), multi-hop nested-WG E2E, post-quantum handshake interop (ML-KEM hybrid, never PQ-only). Phase 3 adds HarmonyOS NEXT + Aurora OS platform shim E2E (the biggest platform risk — real native tunnel-shim work), WASM browser-MASQUE proxy E2E, and a third-party audit + reproducible-build attestation. Each re-arm flips its coverage-ledger cells from `NOT_APPLICABLE/PENDING` to `REQUIRED` mechanically (§6.2).

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8. Execution discipline: determinism, risk-order, four-layer, diary

Determinism (§11.4.50). Every PASS runs N=3 (normal) / N=10 (cycle-validation) against the same artifact MD5 + same rig, identical evidence-hashes; a divergent run is auto-FAIL, no flake escape (`ab_run_n_times`, §5.4).

Risk-ordering (§11.4.132). The suite runs **highest-risk first**: (a) most-recently-worked, (b) historically most-problematic, (c) highest crash/leak likelihood, (d) most-reopened (`reopens_count` from the workable-items DB). For HelixVPN the irreversible-security floor — kill-switch (AC7), default-deny (AC3), RLS (S9), revoke (AC6), key-never-leaves (S2) — runs **before** any convenience/UI test, and only after that set is GREEN with captured evidence does the rest run.

Four-layer per closure (§11.4.4(b)). Every workable item closes only with: (1) a pre-build gate (lint/schema/sh -n/self-validated-analyzer present), (2) a post-build/runtime test on the deployed artifact, (3) a runtime-signature on a **clean** deploy (§11.4.108/.139

— never validate against a stale daemon, defeats B4), (4) a paired §1.1 mutation that makes the gate FAIL (proves the gate is not a bluff).

Reproduce-first + extend (§11.4.146). Every fix authors a RED test that reproduces the defect on the *current broken* artifact (`RED_MODE=1`), flips to a GREEN guard (`RED_MODE=0`) on the fix, then **fans out** across the feature's full case-space (valid/invalid/ boundary/concurrent/chaos) and registers a permanent regression guard (§11.4.135).

Test diary (§11.4.149). Every run appends a row to the item's `test_diary` (date · tested_by ∈ {AI-agent, HelixQA, Operator} · result · observations · evidence_path · feature_class). The schema constraint **rejects a PASS row with an empty evidence_path** — a diary-layer bluff is impossible.

```
CREATE TABLE IF NOT EXISTS test_diary (       -- §11.4.149, git-tracked
    diary_id     INTEGER PRIMARY KEY,
    atm_id       TEXT NOT NULL REFERENCES items(atm_id),
    date_time    TEXT NOT NULL,                  -- ISO-8601 UTC
    tested_by    TEXT NOT NULL CHECK (tested_by IN ('AI-agent','HelixQA','Operator','User')),
    result       TEXT NOT NULL CHECK (result IN ('PASS','FAIL','SKIP')),
    observations TEXT NOT NULL,
    evidence_path TEXT, feature_class TEXT,
    CHECK (result <> 'PASS' OR (evidence_path IS NOT NULL AND length(evidence_path) > 0))  -- §11.4.69
);
```

Operator-escape ⇒ bluff-audit (§11.4.138). If the operator finds a defect the green suite missed, that is by definition a PASS-bluff: it triggers a systematic-debug pass, a bluff-audit naming the exact assertion that should have caught it (`file:line`), a permanent regression guard in the same commit, and the audit committed under `docs/research/<scope>/<defect>_bluff_audit/`.

---

9. The local gate harness (Makefile + git hooks, §11.4.75/.156)

All enforcement is **local** (§11.4.156 — no active remote CI). The seam:

```
# Makefile — the local gate harness (excerpt; full set in [05])
gen:        ; buf generate && flutter_rust_bridge_codegen generate     # proto + FFI
build:      ; cargo build && go build ./... && melos run build
test:       ; cargo test && go test -race ./helix-go/... && melos run test  # boots infra via containers (§5.2)
```

```
e2e:        ; sudo rig/netns_up.sh && rig/test_reach.sh pass &&
        rig/test_reach.sh deny ; sudo rig/netns_down.sh
sec:        ; rig/killswitch_drop.sh && bash scripts/
        secret_audit.sh     # §5.7
chaos:      ; bash scripts/
        stress_chaos.sh                                 # §5.9
bench:      ; bash bench.sh
        3                                               # N=3
        deterministic (§5.13)
qa:         ; go run ./submodules/helix_qa/cmd/orchestrator --
        suite mvp-dod  # 8 ACs, captured evidence (§5.6)
coverage:   ; go run ./tools/coverage_ledger && docs_chain sync --
        context .docs_chain/contexts/helixvpn_spec.yaml
docs-verify:; docs_chain verify --context .docs_chain/contexts/
        helixvpn_spec.yaml  # deterministic gate (§11.4.106)
```

Git-hook layers (§11.4.75): `pre-commit` runs UNIT + `sh -n` + schema-lint + no-log lint + mutation-residue scan (§11.4.84); `pre-push` runs INT (containers boots PG+Redis) + the SEC subset + propagation gates; `post-commit` auto-exports HTML/PDF/DOCX siblings (§11.4.65/.153). The `qa.yml.disabled-local-only` file is preserved but inert (§11.4.75 Layer-5). Long suites run backgrounded (`nohup … & disown`, §11.4.89) so the main work stream is never blocked.

Container-test infra is rootless (§11.4.161). `containers/pkg/boot` runs Podman rootless; no `sudo`, no rootful Docker — the only `sudo` in the whole harness is `rig/netns_up.sh` (network-namespace creation needs `CAP_NET_ADMIN`), and that is a documented, scoped exception, never a container-management escalation.

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10. Open decisions surfaced for QA

Per §11.4.66, decisions the QA layer touches but does not silently resolve:

#	Decision	Options	Recommendation
QA-D1	Edge language (G4) drives which RACE toolchain is canonical	Rust (loom+tsan) vs Go (`-race`)	**Rust** — loom's exhaustive interleaving is stronger for the verdict-swap hot path [04_P0 G4]; revisit on the G4 CSV.
QA-D2	DDoS suite timing	author-now-parked vs defer-authoring	**Author-now-parked** — build the bank in MVP so it re-arms mechanically in Phase 2; the seam (rate-limit,

#	Decision	Options	Recommendation
			fail-static) exists now [04 §T9].
QA-D3	iOS MEM gate on CI-less hosts	real-device-required vs simulator-allowed	**Real-device-required** — the jetsam ceiling does not reproduce in the simulator; no device ⇒ honest `SKIP`: `hardware_not_present`, never a sim PASS (§11.4.3).
QA-D4	Vision verdict authority	trust harness verdict vs independent `vision_engine` re-read	**Independent re-read** — `vision_engine` re-reads the MP4 and cross-checks the core FSM (defeats B3); the harness's claim is never the verdict (§11.4.165).
QA-D5	Challenge engine vs raw test exit codes for DoD	exit-code vs `challenges`-scored	**`challenges`-scored** — the engine re-reads the evidence artifact, structurally defeating B1–B3; exit codes alone are bluffable (§11.4.27).

Sources

- **[04_P0]** `04_VPN_CLD/HelixVPN-Phase0-Spike.md` — netns+nftables+netem rig, G1–G6 gates, `bench.sh`, surviving interfaces, iOS memory gate.
- **[04_P1]** `04_VPN_CLD/HelixVPN-Phase1-MVP.md` — MVP DoD (AC1–AC9), SLOs, no-log schema-lint, RLS, coordinator/ WatchNetworkMap.
- **[04_ARCH]** `04_VPN_CLD/HelixVPN-Architecture-Refined.md` — three roles, transport stack, security invariants.
- **[04_UI]** `04_VPN_CLD/HelixVPN-helix-ui-Flutter.md` — Flutter flavors, design system, UI/golden test surface.
- **[05_YBO]** founding constraints + mandated stack. **[02_QWN] [01_DSK] [11_MST] [10_KMI] [07_GMI]** — LLM analyses (transport plurality, IPAM/D4 solvers, asymmetric per-leg transport).
- **[SYNTHESIS]** `v09-research/_SYNTHESIS.md` — settled floor, key decisions D1–D7, phased structure, ecosystem-integration gaps.
- **[01] [02] [03] [04] [05] [06] [07]** — companion `final/` documents (data plane, control plane, client core+UI, security/ PKI, repo+ecosystem, Phase-0 WBS, Phase-1 WBS).

- External facts: **[research-masque]** RFC 9297 (HTTP Datagrams)/9298 (CONNECT-UDP)/9221 (unreliable datagram); **[research-boringtun]**, **[research-quinn]** (Rust QUIC), **[research-wireguard]** (Noise IK).
- Constitution: §11.4.169 (test-type mandate), §11.4.27 (no-fakes/100% type coverage), §11.4.5/.69/.107 (captured evidence / sink-side / liveness), §11.4.85 (stress+chaos), §11.4.50 (determinism), §11.4.132 (risk-order), §11.4.4(b)/.108/.139/.146/.135 (four-layer / runtime-signature / reproduce-first / regression guard), §11.4.149 (test diary), §11.4.25/.52/.153 (coverage ledger), §11.4.76/.161 (containers/ rootless), §11.4.156 (no active remote CI), §11.4.75 (local hook layers), §11.4.159/.163/.165 (recorded evidence / media validation / independent verification), §11.4.93/.95 (workable-items DB).